ANDREW KIPKOECH

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EDUCATION

Texas A&M University – M.S. Computer Science *Expected May 2027*

College Station, TX | Coursework: Computer Architecture, Computer Networks, Systems Security, Machine Learning, Multi-Robot Systems

Methodist University – B.S. Computer Science *May 2022*

Fayetteville, NC | GPA: 3.63

TECHNICAL SKILLS

Languages: C/C++, Python, Java, Go

AI / ML & GPU: PyTorch, HuggingFace Transformers, PEFT/LoRA, TensorFlow, Scikit-learn, FAISS (RAG), model quantization, NVIDIA Jetson Nano

Systems & Networking: Linux, raw sockets/TCP-IP, multithreading, distributed systems, computer architecture & cache simulation (zsim)

Cloud, Tools & Security: AWS (AppConfig, Lambda), Git, CI/CD (Jenkins), SLURM (HPC scheduling), GDB, AFL++, angr

PROFESSIONAL EXPERIENCE

Visa – Software Engineer *Jun 2022 – Jun 2025*

Austin, TX

- Integrated Apache Kafka into core transaction-processing services, reducing latency and improving scalability for high-volume transaction messaging.
- Built a Fraud Map Insight solution combining TensorFlow/Scikit-learn fraud-detection models, NLP-driven news analysis, and real-time interactive visualization.
- Designed and secured REST API endpoints; automated dependency-vulnerability remediation across CI/CD pipelines, increasing developer productivity by 30%.

Amazon – Software Development Engineer Intern *Jun 2021 – Aug 2021*

Seattle, WA

- Built a Dynamic Configuration system enabling service configs to be stored in and deployed from AWS AppConfig without restarts; implemented in Java and validated with end-to-end integration tests.

FEATURED PROJECTS

Edge-Clinic – LLM Fine-Tuning for Edge Deployment (Texas A&M)

- Fine-tuned T5-small with LoRA (rank=16, alpha=32) using PyTorch, HuggingFace Transformers, and PEFT for clinical NLP on a low-resource Kenyan clinical-vignette dataset.
- Built a retrieval-augmented generation (RAG) pipeline with FAISS that outperformed baseline and plain-LoRA models on ROUGE; applied INT8 quantization to prepare the model for deployment on an NVIDIA Jetson Nano.

Security Research – HackerOne Bug Bounty & Independent Vulnerability Research

- Discovered and reported a plaintext session-cookie storage flaw and an unprotected Content Provider vulnerability in the Wikimedia Commons Android app.
- Fuzzed the libexpat XML parser with AFL++ and AddressSanitizer and documented an API contract violation through manual code audit.
- Built a ptrace-based debugger to intercept credentials from a live process, and used angr symbolic execution plus AFL++ fuzzing to defeat anti-analysis binaries.

Systems & Computer Architecture

- Implemented a Locality-Aware Cache (LAC) replacement policy in zsim; benchmarked against LRU/LFU across SPEC CPU2006 and PARSEC workloads.
- Built a parallel traceroute tool in C++ (raw ICMP sockets, async DNS resolution, dynamic RTO estimation) and a flow-controlled, pipelined UDP sender with RTT estimation.

ADDITIONAL EXPERIENCE

Methodist University – Math/CS Teaching Assistant & Student Leader *Aug 2020 – May 2022*

Fayetteville, NC

- Tutored students on problem sets and debugging, and supervised CS/Math department TAs – adapting explanations of technical concepts for students at different levels.